CHE 45000 – Design and Analysis of Process Systems Spring 2026 Public Syllabus

Course Information and Description

a. **Catalog description:** Use of chemical engineering process and product synthesis methods and concepts, detailed design of unit operation equipment, the economics of chemical plants, and flow sheet optimization methods. Synthesize, develop, and evaluate a preliminary design of a chemical process that meets market requirements for a specific product. Analysis of design alternatives using case studies and optimization methods.
b. **Prerequisites:** CHE 30600 (Design of Staged Separation Processes); CHE 37800 (Heat and Mass Transfer); CHE 43500 (Chemical Engineering Laboratory) (concurrent)
c. **Course Status:** Required
d. **Credits:** 4
e. **Contact Hours:**
   Lecture 3 times per week for 50 minutes
   Laboratory (computer and project) once a week for 110 minutes (including 20-minute team meeting with faculty mentor)

Instructor

J. J. Siirola, Instructor and Faculty Mentor
C. M. Masuku, Faculty Mentor

Textbook

a. **Required:**
   Chemical Engineering Design - Principles, Practice, and Economics of Plant and Process Design, 3rd Edition, Towler, Sinnott, 2021
b. **Other supplemental materials:**
   Learn Aspen Plus in 24 Hours, 2nd ed., Adams, 2021 (Recommended)
   Product and Process Design Principles, Seider, Lewin, Seader, Widagdo, Gani, Ng, 2016
   Analysis, Synthesis, and Design of Chemical Processes, Turton, Bailie, Whiting, Shaeiwitz, Battacharyya, 2012
   Systematic Methods of Chemical Process Design, Biegler, Grossmann, Westerberg, 1997
   Conceptual Design of Chemical Processes, Douglas, 1988
   Plant Design and Economics for Chemical Engineers, Peters, Timmerhaus, 1991
   Chemical Engineers' Handbook, 8th Edition, Perry, Green, 2007

Towler is the required text. Adams is an optional Aspen reference. Seider, Turton, Biegler, Douglas, and Peters are supplementary contemporary texts which provide many insightful alternative perspectives on process design. *Perry's Chemical Engineers' Handbook (8th ed.)* is an invaluable reference for this course and all chemical engineering practice. Familiarity with material in this resource is assumed.

Specific Goals for the Course

Apply systematic strategies for synthesizing chemical process designs that involve conventional unit operations
Describe both the difference between steady state and batch chemical processes and the implications on their design and operation
Demonstrate the ability to obtain reliable information, including through the use of engineering judgement, on industrial chemical processes, process operating parameters,

equipment costs, cost of chemicals and materials, and associated safety and environmental hazards
- Describe the role of physical property estimates on process design and be able to use appropriate physical property estimation methods in unit operations design
- Estimate the capital and operating cost of a process and to assess its profitability
- Use process simulators and other tools to perform detailed analyses for the successful design of a chemical plant
- Communicate effectively project progress and final results in a professional manner both orally and in writing to a wide range of audiences
- Work effectively in a team to execute open-ended design projects with time-bound deliverables in a professional and ethical manner, including different team roles

Student Outcomes Addressed by the Course

- Apply principles of engineering, science, and mathematics to solve complex chemical engineering problems
- Apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors
- Communicate effectively with a range of audiences
- Recognize ethical and professional responsibilities in chemical engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts
- Function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives
- Develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions
- Acquire and apply new knowledge as needed, using appropriate learning strategies

Course Content

- Chemical process introduction and history
- Design representations
- Design paradigm
- Process synthesis
- Process intensification
- Material and energy balances
- Heat integration
- Distillation sequences; azeotropic and extractive distillation; nonideal distillation scheme synthesis.
- Equipment design and capital costing
- Time value of money; net present value
- Economic optimization
- Batch processes
- Plantwide control
- Environmental protection
- Health, personnel safety, process safety
- Ethics, professionalism, and licensure

Grading Policy

- Class, Laboratory, and Mentoring Meeting Participation 5%

Homework (Nine Weekly Individual Assignments)	20%
Laboratory (Individual Simulation Tool Mastery)	15%
Project #1 - Team Technology Assessment Report	10%
Project #2 - Team Equipment Design Report	15%
Project #3 - Team Progress Report 1	5%
Project #3 - Team Progress Report 2	5%
Project #3 - Team Process Design Presentation	10%
Project #3 - Team Process Design Final Report	15%
Bonus Activities	Up to +10%
Bonus Homework	Up to +6%
Bonus Projects	Up to +10%

There will be no final examination.

Course grades are based on a straight scale: 90%+ is an A, 80%+ is a B, 70%+ is a C, 60%+ is a D, and below 60% is an F. Pluses and minuses may be awarded within each percentage decile. Bonus points may be assigned for completely optional individual initiative activities that provide alternative and supplemental course knowledge.

Homework Assignments

Homework is designed primarily to help students learn course material, develop needed engineering skills, and develop capabilities needed for course projects. Each homework assignment will be graded on a 5, 10, or 15-point scale depending on complexity. Bonus homework opportunities are generally worth up to 3 additional points. Homework is due by the start of the lecture on the due date with a cover sheet with the student's name, laboratory section, homework number, and date of submission. Homework will be individual and will be graded and returned as quickly as possible at a subsequent laboratory session. Solutions may be available subsequently on the course Brightspace.

Computer Design Laboratory

The FRNY PC facility will be used for teaching the laboratory component of the course. The lab component involves a progression of computer exercises to give the student some practical experience with process simulation tools used in industry and to develop facility working in teams. The laboratory is important since students will use these computer tools extensively for the detailed equipment design and process design projects (Projects #2 and #3).

Attendance is required in the assigned laboratory section. During the first half of the course, students will complete a series of exercises leading up to the laboratory mastery examination. Laboratory exercises may be individual or team as instructed, but the mastery examination will be individual and will be graded. During the second half of the course, students will use the laboratory time for team project work. A portion of the laboratory time will also be used for meetings with the faculty mentor. The tentative schedule for the first part of the laboratory is:

Week 1	Team formation; working in teams; basic ASPEN review
Week 2	ASPEN design specs; sensitivity; optimization
Week 3	Simple process flowsheet simulation
Week 4	Analysis of automatized process flowsheet simulation

Week 5	Process flowsheet optimization
Week 6	Integrated/realistic process flowsheet simulation
Week 7	ASPEN Energy Analyzer: Heat integration of flowsheets
Week 8	ASPEN Economic Analyzer: Process economic evaluation
Week 9	Process simulation tool mastery examination

Course Projects

There will be three course projects, all completed in teams. Project #1 (3.5 weeks) is a literature review and assessment of alternative technologies for a specific function. Project #2 (3 weeks) is an exercise in detailed equipment design and optimization. Project #3 (7 weeks) is the major course process design project intended to be a capstone experience requiring integration of information and skills acquired throughout the entire curriculum. It will involve the conceptual design and economic evaluation of a process to perform a particular function. During the project, two brief written progress reports will be submitted to faculty mentors. The project will conclude with an oral PowerPoint presentation and a written final report. Team presentations will likely be attended by industrial and other guests who will contribute to the assessment of the project solutions and oral presentation quality.

During the weekly laboratory period there will be regularly scheduled nominally 20-minute weekly meetings of each team with a faculty mentor who will assess team progress and offer project advice. Several times throughout the course, each student will rate the contribution of other members of their team to the group. All projects and presentations will receive a score according to rubrics published in advance. The peer evaluations along with the faculty mentor assessment of individual contributions then will be used to adjust the project grade assigned to each individual team member as appropriate.

Bonus Opportunities

Throughout the course, there will be a number of completely optional activities to broaden the understanding of the systems design and analysis process incentivized by the opportunity to earn additional credits toward the final grade. These will be announced in class throughout the semester. Bonus opportunities may take several forms including for example professional development activities, seminar reports, special homework problems, literature study, plant trips, short projects, etc.

Grade Revisions

Invariably, questions arise concerning the fairness of a grade on a homework, laboratory, or project. The following policy applies to re-grading: (i) the entire work is subject to revision in the grade – not just a single item; (ii) the request to re-grade must be made within one week of the date that the work was returned. If the re-grading applies to laboratory or homework assignments, the request should be made to the TA. If the re-grading applies to projects, the request should be made to the course instructor.

CHE 450 Brightspace

Brightspace learning management system will be set up for CHE 450. We will post important class announcements, homework and project assignments, homework solutions, lecture notes, project definition updates, hints, responses to questions, etc. Students are responsible for awareness of all posts.

Academic Honesty

Students are individually responsible for each homework and the laboratory mastery examination. Cheating will not be tolerated. While discussions of homework among classmates are to be expected, students are responsible for submitting their own work on all individual assignments. Copying without attribution is plagiarism and is considered a form of cheating.

Accommodation

Purdue University strives to make learning experiences as assessable as possible. If students anticipate or experience physical or academic barriers based on disability, they are welcome to let the instructor know so that options may be discussed. Students are also encouraged to contact the Disability Resource Center at drc@purdue.edu or by phone at 765-494-1247.

In addition to the University policy, the Davidson School of Chemical Engineering has established procedures for students seeking accommodations. These can be found online at the ChE Undergrad Office website. Only those accommodation requests that conform to both University and ChE policy guidelines will be implemented.